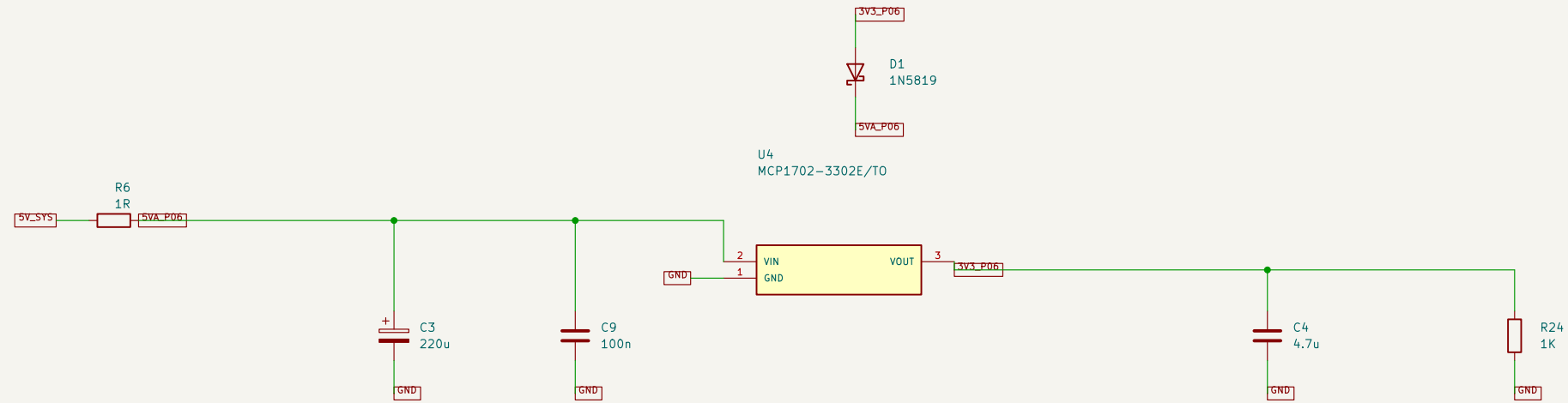
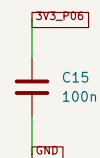
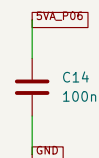
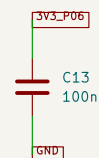
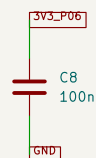
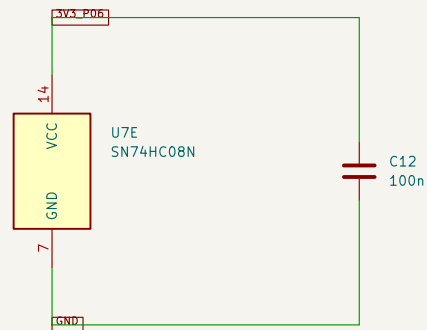
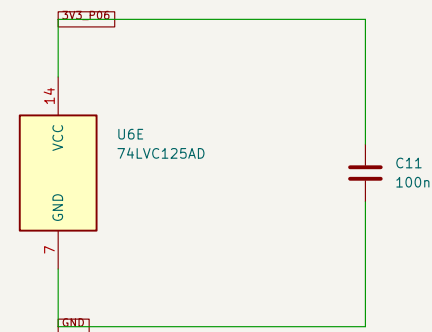
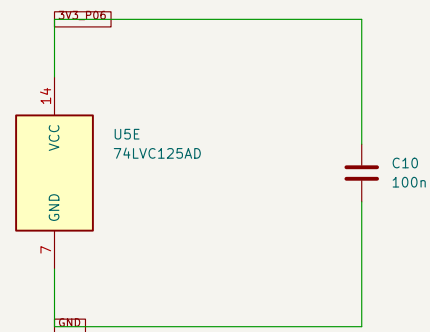
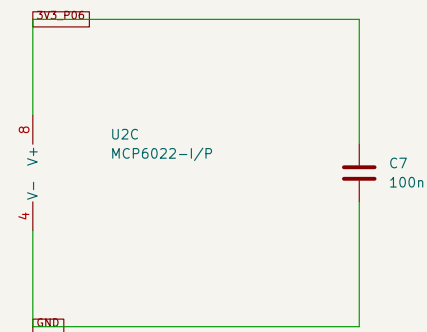


P06-R2 / 01 ZASILANIE LOKALNE



J_BP: 5V_SYS (piny 10/12) zasila plytkę. 3V3_I0 (J_BP.14) idzie tylko na kolek J_SV2. Logika pracuje z lokalnego 3V3_P06.



FORCE

Plik: FORCE.kicad_sch

ANA

Plik: ANA.kicad_sch

DIG

Plik: DIG.kicad_sch

READY

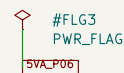
Plik: READY.klcad_sch

CONNECT

Pluk: CONNECT.kicad_sch

SERWIS

Pluk: SERWIS.kicad sch



D1: anoda 3V3_P06, katoda 5VA_P06. Budzet MEASURE: 180 mA, wiekszość poboru przez R21 obciążający styk. R6 1R/1W leżący; C3 220u (1.10); impuls ładowania ok. 3 mJ.

Inline current measurement; format S1 (2/3, level 4, S1–S2); schematic only, hardware not tested

Prototype / schematic + PCB review

Sheet: /
File: P06.kicad_sch

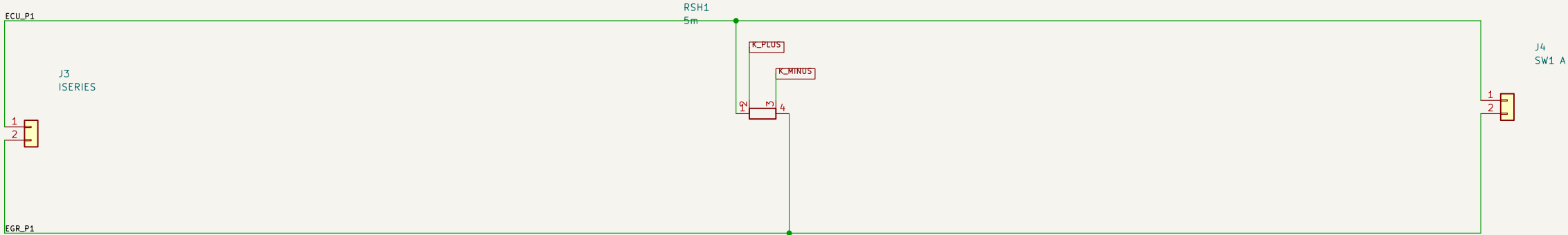
Title: EGRLab P06 I-LOGGER

Size: A3	Date: 2026-10-01
KiCad E.D.A. 10.0.6	

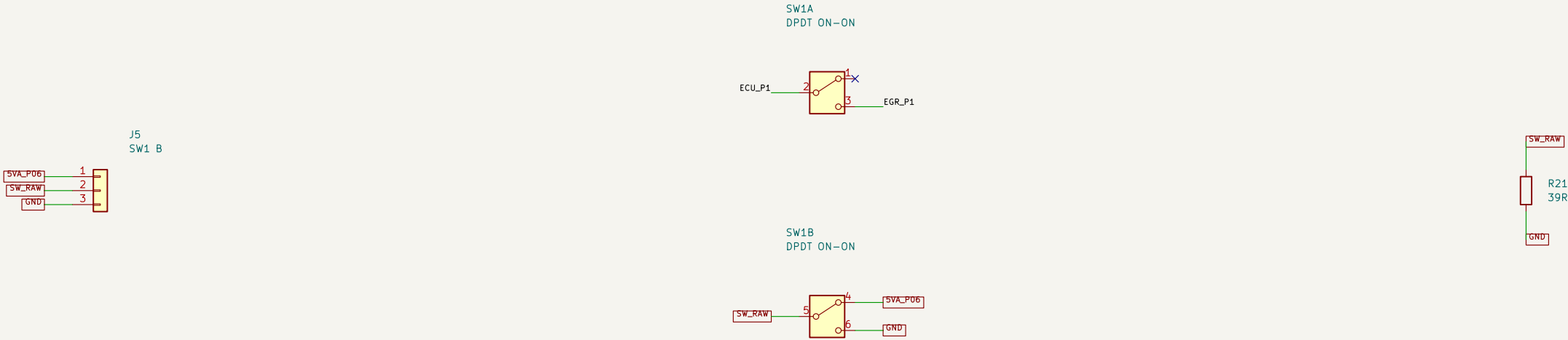
Date: 2026-10-01

Rev: P06-R2
Id: 1/7

RSH1 5 m0hm 2512 Kelvin (pady 1/4 pradowe, 2/3 pomiarowe) stale w torze ECU_P1 -> EGR_P1. Brak przełącznika szeregowego i połączenia mocy z GND.



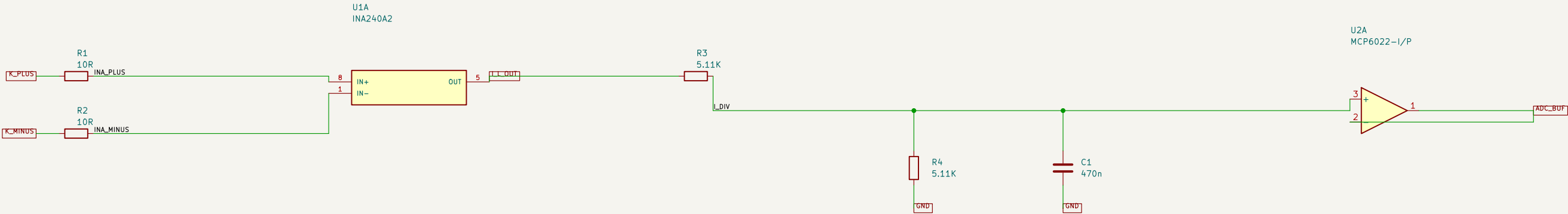
BYPASS: SW1 2-3 + 5-6. MEASURE: SW1 2-1 (NC) + 5-4. COM to zaciski 2 i 5, nie 1 i 4.



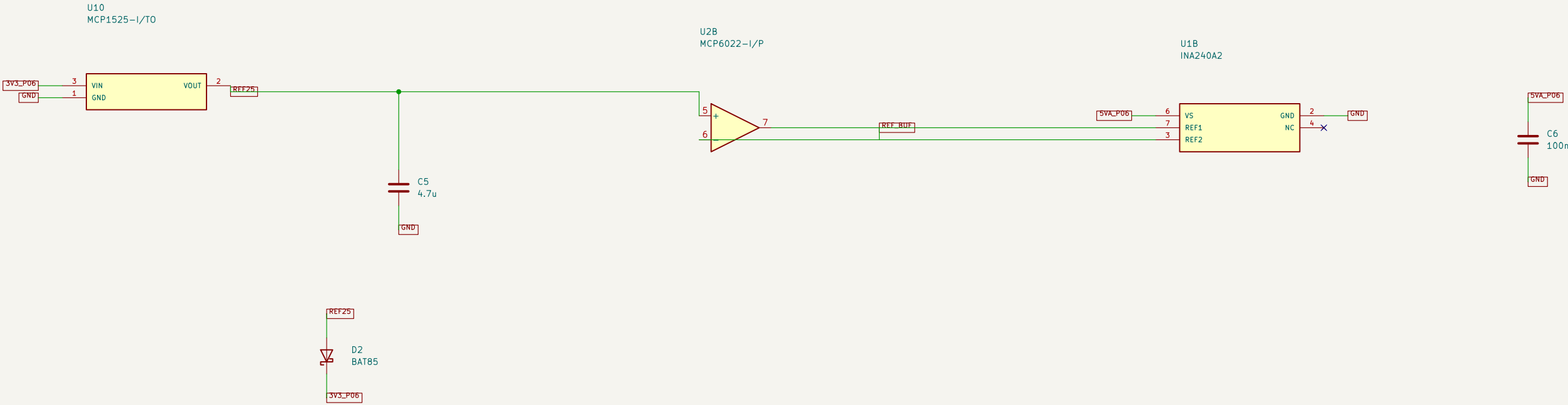
SW1 na panelu (DPDT ON-ON >=10 A DC, oczka). J4.1 -> SW1.2; J4.2 -> SW1.3, J5.1 -> SW1.4; J5.2 -> SW1.5; J5.3 -> SW1.6. J3/J4/J5 przy brzegu x=0.

R21: 39R PR02 2W lezacy, prad styku ok. 0.13 A (0.71 W przy 5.25 V). Goracy element oddalony od boczniaka. Nie przełączac przy włączonym zaplonie.

5 mOhm x 50 V/V x 1/2 = 0.125 V/A. ADC: 1.25 V przy 0 A, idealnie 204.8 kodu/A; kalibracja offsetu i nachylenia obowiązkowa.



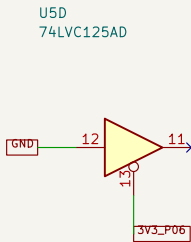
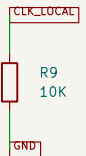
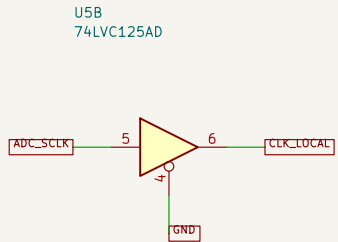
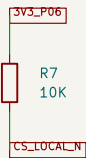
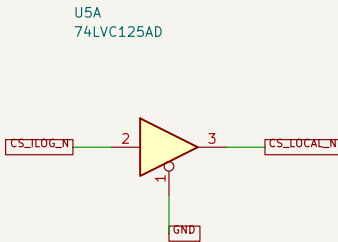
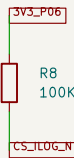
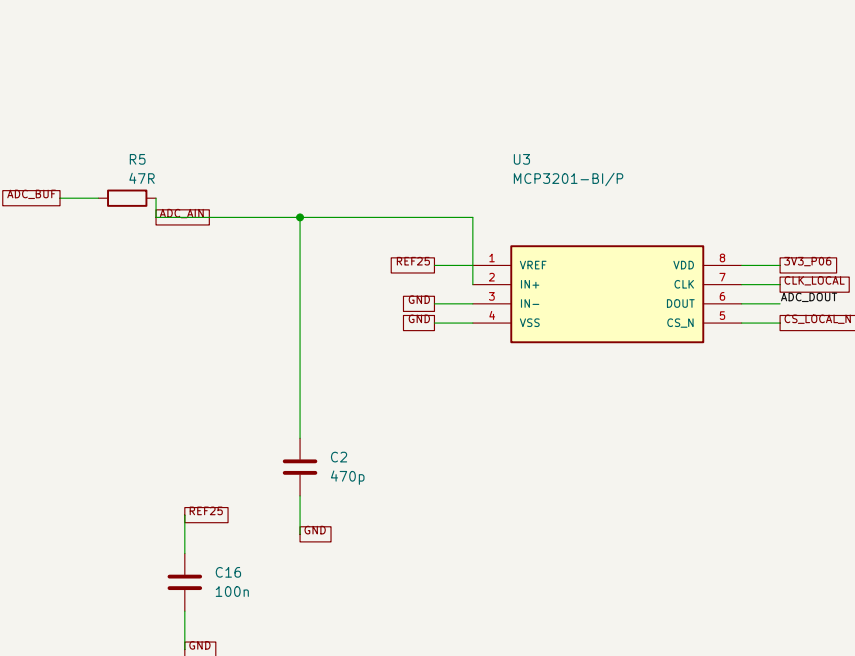
R3/R4 5.11k 0.1% (R3 || R4 = 2.555k); C1 = 470n X7R; fc = 133 Hz, tau = 1.20 ms. Pomiar trendu prądu; nie przebieg pojedynczych impulsów PWM.



U1 SOIC D: 8 IN+, 1 IN–, 5 OUT, 6 VS, 2 GND, 7 REF1, 3 REF2, 4 NC. Oba REF sterowane wspólnym buforem.

U10 T092: 1 GND, 2 VOUT, 3 VIN. C5 4.7uF X7R 1206 przy U10; R1/R2 10R 0.1% identyczne, ścieżki Kelvin lokalne.

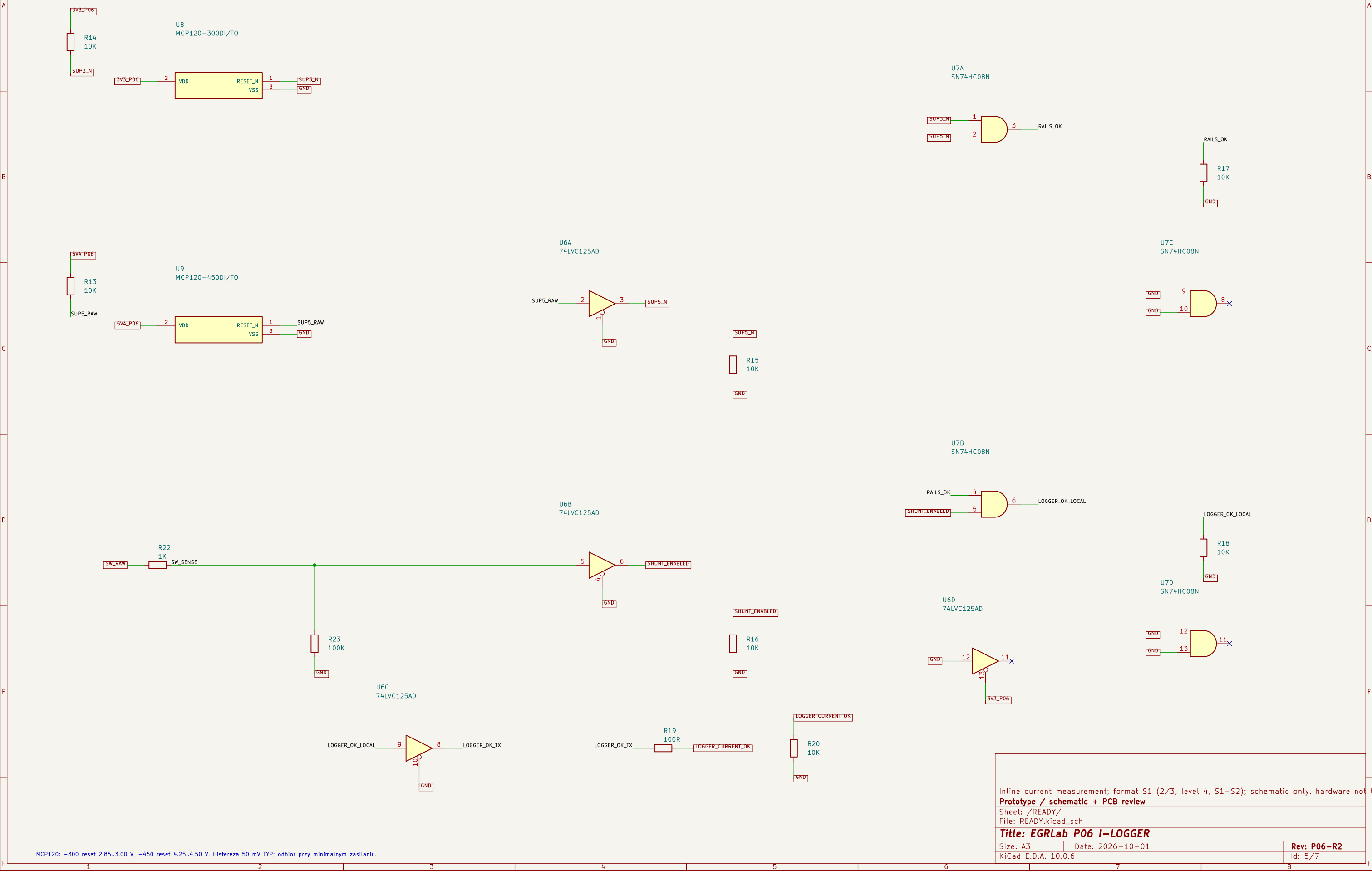
SPI mode 0, 500 kHz, 16 zegarow. raw = (word >> 1) & 0x0FFF. DOUT odlaczony od wspolnej magistrali gdy CS_LOCAL_N=1.



2 kS/s z firmware v6.1. Probka niesymultaniczna wzgledem AD7606B; zachowac timestamp/skew. Nie wyprowadzac I2C do analogu.

U5/U6 tylko Nexperia 74LVC125AD z loff. HC125 nie jest zamiennikiem. SOIC/SO14 raster 1.27 mm.

LOGGER_CURRENT_OK = SUP3_N & SUP5_N & SHUNT_ENABLED; BYPASS daje LOW. READY nie jest dowodem kalibracji ani sprawnosci bocznika.



Inline current measurement; format S1 (2/3, level 4, S1-S2); schematic only, hardware not tested

Prototype / schematic + PCB review

Sheet: /READY/
File: READY.kicad_sch

Title: EGRLab P06 I-LOGGER

Size: A3

Date: 2026-10-01

Rev: P06-R2

KiCad E.D.A. 10.0.6

Id: 5/7

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P06–R2 / 06

ZŁACZE J_BP DO P12 (KRAWĘDZ A)

J_BP: katowe obudowane IDC 2x8 na krawędzi A, slot S2 (srodek x=80.0 mm), pin 1 od strony mniejszego x. Tasma ok. 30 mm do P12.

ADC_SCLK/ADC_DOUTA na pinach 2/4 jak J_BP2 płytek P03 R6 i P05 R3. Nieparzyste GND, 16 GND (rezerwa). Zastępuje J1 LV06 i J2 ILOG z R1.

J_BP

J_BP

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

GND

ADC_SCLK

ADC_DOUTA

CS_ILOG_N

LOGGER_CURRENT_OK

SV_SYS

SV_SYS

SV3_I0

GND

3V3_I0 (pin 14) nie ma odbiorcy na P06 (jak LV06.3 -> TP4 w R1): tylko kolek J_SV2.6 przez 1k.

Inline current measurement; format S1 (2/3, level 4, S1–S2); schematic only, hardware not tested

Prototype / schematic + PCB review

Sheet: /CONNECT/
File: CONNECT.kicad_sch

Title: EGRLab P06 I–LOGGER

Size: A3

Date: 2026–10–01

Rev: P06–R2

KiCad E.D.A. 10.0.6

Id: 6/7

1

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8

J_SV1 (slot S1, x=10..43): wezly analogowe przez 10k, GND na 1, 4, 7. J_SV2 (slot S2, x=63.5..96.5): szyny i logika, GND na 1, 5, 13.

Rezystor przy wezle: 1k szyny/logika, 10k wezly analogowe i SUP3_N/SUP5_N. Szyna tylko obok GND, innej szyny albo linii 10k. Numeracja zawsze od pinu 1.

